3.Operators and Expressions

Task1:

Code:

```
import Foundation

let itemPrice = 450.0

let quantity = 3

let subtotal = itemPrice * Double(quantity)

let hasCoupon = true

let discount = (hasCoupon && subtotal > 1000) ? subtotal * 0.10 : 0.0

let discountedAmount = subtotal - discount

let gst = discountedAmount * 0.18

let total = discountedAmount + gst

let share = total / 4.0

print("Subtotal: ₹\(subtotal)")

print("Discount: ₹\(discount)")

print("GST: ₹\(gst)")

print("Total Bill: ₹\(total)")

print("Each Person's Share: ₹\(share)")

let amount = 1234

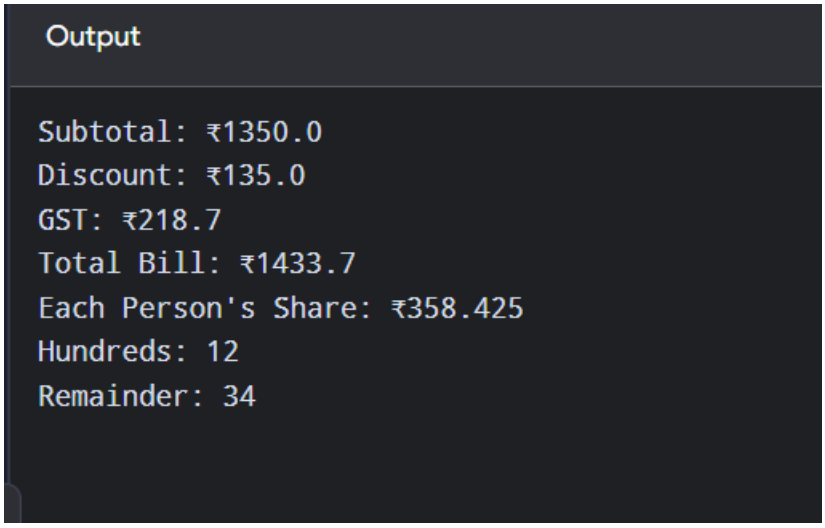
let hundreds = amount / 100

let remainder = amount % 100

print("Hundreds: \(hundreds)")

print("Remainder: \(remainder)")
```

Output:



Task2:

Code:

```
let year1 = 1900

let year2 = 2000

let year3 = 2024

let year4 = 2026

let leap1 = (year1 % 4 == 0 && year1 % 100 != 0) || year1 % 400 == 0

let leap2 = (year2 % 4 == 0 && year2 % 100 != 0) || year2 % 400 == 0

let leap3 = (year3 % 4 == 0 && year3 % 100 != 0) || year3 % 400 == 0

let leap4 = (year4 % 4 == 0 && year4 % 100 != 0) || year4 % 400 == 0

print("1900 is a leap year: \(leap1)")

print("2000 is a leap year: \(leap2)")

print("2024 is a leap year: \(leap3)")

print("2026 is a leap year: \(leap4)")

let number = 50

let inRange = (1...100).contains(number)
```

```
print("Number is between 1 and 100: \(inRange)")

let n = 24

let result = (n % 2 == 0) && (n % 10 != 0)

print("Even and not a multiple of 10: \(result)")

let nickname: String? = nil

let displayName = nickname ?? "Guest"

print("Nickname: \(displayName)")
```

Output:

Output

```
1900 is a leap year: false
2000 is a leap year: true
2024 is a leap year: true
2026 is a leap year: false
Number is between 1 and 100: true
Even and not a multiple of 10: true
Nickname: Guest
```

Task3:

Code:

```
print(7 / 2)

print(7 % 2)

print(-7 % 3)

print(7.0 / 2)

print(10 - 2 * 3 + 4 / 2)

print(5 > 3 && 2 > 4 || true)

print(1 << 3)
```

```
print(6 & 3, 6 | 3, 6 ^ 3)

let x: UInt8 = 255

print(x &+ 1)
```

Output:

